

Geometry-Topology Qualifying Examination — September 25, 2002

Problem 1

Suppose $P(x, y, z)$, $Q(x, y, z)$, and $R(x, y, z)$ are C^∞ functions on $\mathbb{R}^3$ which vanish identically if $|x| \geq 5$, $|y| \geq 5$, or $|z| \geq 5$. Prove that the volume integral

$$\int_{-6}^{+6} \int_{-6}^{+6} \int_{-6}^{+6} d(Pdy \wedge dz + Qdx \wedge dz + Rdx \wedge dy) = 0.$$

(Do this directly, not by quoting Stokes' Theorem: this is a special case of the proof of Stokes' Theorem!)

Solution.

$$d(Pdy \wedge dz + Qdx \wedge dz + Rdx \wedge dy) = P_x dx \wedge dy \wedge dz + Q_y dy \wedge dx \wedge dz + R_z dz \wedge dx \wedge dy$$

$$\int_{[0,6]^2} \int_{-6}^6 P_x dx \wedge dy \wedge dz = \int_{-6}^6 (P(6) - P(-6)) dy \wedge dz = 0.$$

Similarly for Q and R . Thus the total integral is 0.

Problem 2 [TODO]

Suppose that $V = P(x, y, z) \frac{\partial}{\partial x} + Q(x, y, z) \frac{\partial}{\partial y} + R(x, y, z) \frac{\partial}{\partial z}$ is a C^∞ vector field on $\mathbb{R}^3$ with $V \neq \vec{0}$ at the origin. Find a necessary and sufficient condition for there to exist a C^∞ function $\lambda(x, y, z)$ in some neighborhood of the origin such that λV is the gradient of a C^∞ function on the neighborhood.

Proof:

□

Problem 3

Let $T_t : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the right-hand rule rotation around the positive z -axis by t degrees and $S_s : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the right-hand-rule rotation around the positive x -axis by t degrees.

- Find the infinitesimal generators of the flows T_t and S_t , i.e., the vector fields X and Y , respectively, on $\mathbb{R}^3$ whose flows are $\{T_t\}$ and $\{S_t\}$.
- Compute the commutator $T_{-t} \circ S_{-t} \circ T_t \circ S_t$.
- Compare the result of (b) (lowest order non-identically zero term) with the Lie bracket $[X, Y]$.

Proof: We have that:

$$T_t(\vec{x}) = \begin{pmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{pmatrix} \vec{x}$$

$$S_t(\vec{x}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(t) & -\sin(t) \\ 0 & \sin(t) & \cos(t) \end{pmatrix} \vec{x}$$

1.

$$\begin{aligned} X(\vec{x}_0) &= \frac{\partial}{\partial t} [T_t(x)]_{t=0} \\ &= \frac{\partial}{\partial t} \left[\begin{pmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{pmatrix} \vec{x} \right]_{t=0} (x) \\ &= \begin{pmatrix} -\sin(t) & -\cos(t) & 0 \\ \cos(t) & -\sin(t) & 0 \\ 0 & 0 & 0 \end{pmatrix}_{t=0} \vec{x} \\ &= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = A\vec{x} \\ &= -y\partial_x + x\partial_y \\ Y(\vec{x}_0) &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \vec{x} = B\vec{x} = -z\partial_y + y\partial_z \end{aligned}$$

2. We have

$$\begin{aligned} T_t &= e^{tA} = I + At + \frac{1}{2}t^2A^2 + O(t^3) \\ S_t &= e^{tB} = I + Bt + \frac{1}{2}t^2B^2 + O(t^3) \\ T_{-t} &= e^{-tA} = I - At + \frac{1}{2}t^2A^2 + O(t^3) \\ S_{-t} &= e^{-tB} = I - Bt + \frac{1}{2}t^2B^2 + O(t^3) \end{aligned}$$

Thus

$$\begin{aligned} T_{-t}S_{-t}T_tS_t &= I + (AB - A^2 - AB - BA - B^2 + AB + A^2 + B^2)t^2 + O(t^3) \\ &= I + (AB - BA)t^2 + O(t^3) \\ &= I + [A, B]t^2 + O(t^3) \end{aligned}$$

3. Calculating the Lie Bracket:

$$\begin{aligned}
 [X, Y] &= (-y\partial_x + x\partial_y)(-z\partial_y + y\partial_z) - (-z\partial_y + y\partial_z)(-y\partial_x + x\partial_y) \\
 &= yz\partial_x\partial_y - y^2\partial_x\partial_z - xz\partial_y\partial_y + x(\partial_z + y\partial_y\partial_z) - (z(\partial_x + y\partial_y\partial_x) - xz\partial_y\partial_y - y^2\partial_z\partial_x + xy\partial_z\partial_y) \\
 &= x\partial_z - z\partial_x = -z\partial_x + x\partial_z = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = -[A, B] \begin{pmatrix} x \\ y \\ z \end{pmatrix}
 \end{aligned}$$

□

Problem 4

Take as given that a C^∞ 2-form ω on S^2 is of the form $d\theta$ for some C^∞ 1-form θ if and only if $\int_{S^2} \omega = 0$. Use this to show that every C^∞ 2-form Ω on $\mathbb{R}P^2$ has the form $d\Lambda$ for some C^∞ 1-form Λ . (Do not just quote DeRham's Theorem here.)

Proof: Let A be the antipodal map of S^2 . $q : S^2 \rightarrow \mathbb{R}P^2$ standard quotient map, which induces $q^* : \Omega^2(\mathbb{R}P^2) \rightarrow \Omega^2(S^2)$. Note that q^* is an injective map. Note $A^*(q^*(\omega)) = q^*(\omega)$. But note A flips the orientation of S^2 . $\forall \psi \in \Omega^2(\mathbb{R}P^2)$,

$$\begin{aligned}
 I &= \int_{S^2} q^*(\psi) = \int_{S^2} A^*(q^*(\psi)) \\
 &= \int_{A(S^2)} q^*(\psi) = - \int_{S^2} q^*(\psi) \Rightarrow I = 0
 \end{aligned}$$

Thus $q^*(\psi) = d\theta$. $d\theta = A^*(q^*(\psi)) = A^*(d\theta) = d(A^*(\theta))$. Let $\tilde{\theta} = \frac{1}{2}[\theta + A^*(\theta)]$, note $A^*(\tilde{\theta}) = \tilde{\theta}$. Thus $\tilde{\theta}$ factors through $\Omega^1(\mathbb{R}P^2)$. Thus $\psi = d\tilde{\theta}$.

□

Problem 5 [TODO]

- (a) Suppose $F : S^1 \rightarrow \mathbb{R}^3$ is a C^∞ function such that dF is nowhere zero (on S^1). Prove that there is a two-dimensional subspace P of $\mathbb{R}^3$ such that $\pi_P \circ F : S^1 \rightarrow \mathbb{R}^3$ has nowhere vanishing differential, where π_P = orthogonal projection on P .
- (b) Show by example (a picture with explanation is all right) that there is such an F that is also 1 to 1 (injective) but is such that, for all P , $\pi_P \circ F$ fails to be injective.
- (c) Show that if $F : S^1 \rightarrow \mathbb{R}^4$ is C^∞ and injective then there is a three-dimensional subspace H of $\mathbb{R}^4$ such that $\pi_H \circ F$ is injective, where π_H = orthogonal projection on H .

Proof: (a): Since $dF_p \neq 0$ for all $p \in S^1$. Thus dF_p is an injective map. Thus F is an immersion. □

Problem 6

- (a) Suppose $F : S^n \rightarrow S^n$ is fixed-point free (i.e., for all $p \in S^n$, $p \neq F(p)$). Show that F is homotopic to the antipodal map $p \mapsto -p$, $p \in S^n$.

(b) Use part (a) to show that every vector field on (tangent to) S^{2n} , $n = 1, 2, 3, \dots$, vanishes somewhere on S^{2n} (i.e., has a zero).

Proof: (a): Suppose $F : S^n \rightarrow S^n$ is fixed point free. Let $H : S^n \times I \rightarrow S^n$ where

$$H(x, t) = \frac{F(x)t - (1-t)x}{\|F(x)t - (1-t)x\|}$$

Note since $F(x) \neq x$ for all $x \in S^n$, $\nexists t \in [0, 1]$ such that $F(x)t - (1-t)x = 0$. Thus H is a homotopy between $x \rightarrow -x$ and F .

(b): Let X be a vector field. Assume the contrary, $\forall p \in S^{2n}$ we have that $X(p) \neq p$. Since S^{2n} is compact and $\|X(\cdot)\|$ is smooth, $\exists M \in \mathbb{R}$ such that $\|X\|_\infty < M$. Let $\varepsilon > 0$ be small enough such that $\varepsilon M < \pi$. Let $F : S^{2n} \times [0, 1] \rightarrow S^{2n}$ where $F(p, t) = \exp_p(t\varepsilon X(p))$. Thus F is a homotopy between the identity map and $F(\cdot, 1)$ a map with no fixed point which is homotopic to antipodal map. $\deg(\text{id}_{S^{2n}}) = 1 = \deg(F(\cdot, 1)) = \deg([x \rightarrow -x]) = (-1)^{2n-1} = -1$. This is a contradiction. $\square$

Problem 7 [TODO]

- (a) Discuss carefully how to obtain the long exact sequence in homology from a short exact sequence of chain complexes. (Include definitions of the maps in the long exact sequence.)
 (b) If the short exact sequence is

$$0 \rightarrow C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow 0,$$

prove exactness of the long exact sequence at $H_k(C_3)$ [in $\dots H_k(C_2) \rightarrow H_k(C_3) \rightarrow H_{k-1}(C_1) \dots$].

Problem 8

(a) Suppose $F : T^2 \rightarrow T^2$ (where $T^2 = S^1 \times S^1$) is a continuous function such that $F(p) = p$ for some $p \in T^2$ and

$$F_* : \pi_1(T^2, p) \rightarrow \pi_1(T^2, p)$$

is the identity map. Is F necessarily homotopic to the identity map from T^2 to itself?

(b) Is a C^∞ map $F : T^2 \rightarrow T^2$ of degree 1 necessarily homotopic to the identity map of T^2 to itself? Explain/prove your answer.

Solution. (a): $\tilde{F} : T^2 \rightarrow \mathbb{R}^2$ where $\tilde{F}(x, y) = F(x, y) + (1, 1)$. Note $F = q \circ \tilde{F}$. Let H be the standard straight line homotopy from $\text{id} \rightarrow \tilde{F}$. Thus $q \circ H$ is the straight line homotopy from $\text{id} \rightarrow F$.

(b): Take $F : T^2 \rightarrow T^2$ antipodal map $x \rightarrow -x$. The $\deg F = 1$, because F consists of 2 reflections. But its induced map on $\pi_1(T^2, p)$ is $-I$, thus it cannot be homotopic to the identity map on T^2 .

Problem 9

- (a) Discuss the representation of $\mathbb{C}P^n$ as a cell complex.
 (b) Use part (a) to find the homology of $\mathbb{C}P^n$: prove carefully that your calculation is correct.

Solution. (a): You have a 0-cell, 2-cell, ..., $2n$ -cell. To attach the cells, you attach cell $2(n-1)$ to cell $2n$ by attaching the boundary to $2(n-1)$ cell directly.

(b): You have the complex neighborhood of s :

$$0 \rightarrow C_{2n} = \mathbb{Z} \rightarrow 0 \rightarrow C_{2(n-1)} = \mathbb{Z} \rightarrow \dots \xrightarrow{d_3} C_2 = \mathbb{Z} \xrightarrow{d_2} 0 \xrightarrow{d_1} C_0 = \mathbb{Z} \rightarrow 0$$

Thus we have $0 \xrightarrow{d_{2k}} C_{2k} \xrightarrow{d_{2k-1}} 0 \Rightarrow H_{2k}(\mathbb{C}P^n) = \mathbb{Z}$ for $H_{2k-1}(\mathbb{C}P^n) = 0$.

Problem 10

(a) Let X be the space obtained by attaching two discs to S^1 , the first disc being attached by $S^1 = \partial D_1 \rightarrow S^1$ being the 7 times around (counterclockwise) map, e.g., $z \rightarrow z^7$, $|z| = 1$, $z \in \mathbb{C}$, and the second being attached by $S^1 = \partial D_2 \rightarrow S^1$ being the 5 times around map $z \rightarrow z^5$. Find the homology of X .

(b) Can X be made a C^∞ manifold? Why or why not?

Proof: (a): We have this cellular chain complex:

$$0 \rightarrow \mathbb{Z}^2 \xrightarrow{\begin{pmatrix} 7 & 5 \end{pmatrix}} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0.$$

Thus

$$H_0(X) = \mathbb{Z}$$

$$H_1(X) = \frac{\mathbb{Z}}{7\mathbb{Z} + 5\mathbb{Z}} = 0$$

$$H_2(X) = \mathbb{Z} \left\langle \begin{pmatrix} -5 \\ 7 \end{pmatrix} \right\rangle = \mathbb{Z}$$

$$7x + 5y = 0 \Rightarrow 7x = -5y \Rightarrow x = -\frac{5}{7}y$$

(b): X consists of finitely many cells, and hence is compact. Furthermore it has a dimension $d = 2$. Assume the contrary, that X is a 2 dimensional C^∞ manifold. Fix a $p \in S^1 \subset X$. There exists a neighborhood of p , N such that N is diffeomorphic to U in $\mathbb{R}^2$. Because the attaching maps of the disks are $z \rightarrow z^7$ and $z \rightarrow z^5$ respectively, p has exactly 7 preimages on the boundary of disk 1 and 5 preimages on the boundary of disk 2. Therefore the neighborhood N contains an open interval $I \subset S^1$, where the 7 half disks from the first disk and the 5 half disks from the 2nd disk are all glued together at the interval I . Now if we look at $N \setminus I$, it has 12 connected components. Whereas $U \setminus I$, a arc in $\mathbb{R}^2$, has exactly 2 connected components. Thus U and N cannot be homeomorphic hence diffeomorphic. Thus a contradiction occurs and X cannot be a C^∞ manifold.

□

Reflection

I want to review the following topics:

1. Local Degree (Poincare Duality in Hatcher + Differential topology). Broadly degree theory - Hatcher Problems + Differential Topology Perspective.
2. Cellular Homology

3. Integrability and Frobenius Theorem

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Problem 11 [TODO]

Let M be a connected smooth manifold. Show that for any two non-zero tangent vectors v_1 at point x_1 and v_2 at point x_2 , there is a diffeomorphism $\varphi : M \rightarrow M$ such that $\varphi(x_1) = x_2$ and $d\varphi(v_1) = v_2$.

Solution.

Problem 12

Let X and Y be submanifolds of $\mathbb{R}^n$. Prove that for almost every $a \in \mathbb{R}^n$, the translate $X + a$ intersects Y transversely.

Solution. Let $E : X \rightarrow \mathbb{R}^n$ be the embedding of X in to $\mathbb{R}^n$. We will solve this by proving that the map $F : X \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ where $F(x, a) = x + a$ is transverse to Y . Hence by Transversality Stability and Sard's Theorem, for almost every $a \in \mathbb{R}$ we have $F(X, a)$ is transverse to Y .

$$dF_{(x,a)} = (dE_x \ I).$$

Note that $dF_{(x,a)}$ is surjective for all $x \in X$ and $a \in \mathbb{R}$. Thus

$$\text{im } dF_{(x,a)}(T_{(x,a)}(X \times \mathbb{R}^n)) = \mathbb{R}^n \cong T_{x+a}\mathbb{R}^n.$$

Thus F is trivially transverse to Y .

Problem 13

Let $M_{n \times n}(\mathbb{R}) \cong \mathbb{R}^{n^2}$ be the space of $n \times n$ matrices with real coefficients.

- Show that $SL(n, \mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) \mid \det(A) = 1\}$ is a smooth submanifold of $M_{n \times n}(\mathbb{R})$.
- Identify the tangent space to $SL(n, \mathbb{R})$ at the identity matrix I_n .

- Show that $SL(n, \mathbb{R})$ has trivial Euler characteristic. [TODO]

Solution. (a): We will show that the value 1 is a regular value for $\det : M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}$. Then $SL(n, \mathbb{R}) = \det^{-1}(\{1\})$ is a embedded smooth manifold of $M_{n \times n}(\mathbb{R}, n)$ by the regular value theorem. Note that $\det$ is a polynomial in the entries of a given matrix and hence is smooth. Let $A \in SL(n, \mathbb{R})$. Let $N(T)_{i,j}$ is the (i, j) minor. Thus $\forall T \in M_{n \times n}(\mathbb{R})$ we have that

$$\det(T) = \sum_{j=0}^1 (-1)^{i+j} T_{i,j} \det(N(T)_{i,j}).$$

Thus

$$d(\det)_T(E_{ij}) = (-1)^{i+j} \det(N_{i,j}(T)).$$

Since $A \in \mathrm{SL}_n(\mathbb{R})$, there exists at least 1 $\det(N(T)_{i,j}) \neq 0$. Thus $d(\det)_T(E_{i,j}) \neq 0$, and $d(\det)$ is a surjective map since $\ker d(\det) < T_A M_{n \times n}(\mathbb{R})$, thus by rank nullity $\mathrm{im} d(\det) = \mathbb{R}$. Thus A is a regular point of $\det$ function. This is true for all elements of preimage of $\det^{-1}(\{-1\})$, hence -1 is regular value.

(b):

$$T_I(\mathrm{SL}_n(\mathbb{R})) = T_I(f^{-1}(\{1\})) = T_I \ker(d(\det)_I)$$

$\forall X \in M_{n \times n}(\mathbb{R})$, let X_i denote the i -th row and let $g_X(t) := I + tX$ curve.

$$\begin{aligned} \det(g_X(t)) &= \det(e_1 + tX_1, \dots, e_n + tX_n) \\ &= \det(e_1, e_2 + tX_2, \dots, e_n + tX_n) + \det(tX_1, e_2 + tX_2, \dots, e_n + tX_n) \\ &= \det(I) + t \sum_{i=1}^n \det(e_1, \dots, e_{i-1}, X_i, e_i, \dots, e_n) + O(t^2) \end{aligned}$$

Then

$$\begin{aligned} \frac{d}{dt}[\det(g_X(t))]_{t=0} &= \left[\sum_{i=1}^n \det(e_1, \dots, e_{i-1}, X_i, e_i, \dots, e_n) + O(t) \right]_{t=0} \\ &= \sum_{i=1}^n \det(e_1, \dots, e_{i-1}, X_i, e_i, \dots, e_n) = \mathrm{tr}(X) \end{aligned}$$

Thus $\ker d(\det)_I = \{X \in \mathrm{Mat}_{n \times n}(\mathbb{R}) : \mathrm{tr}(X) = 0\}$. Hence $T_I \mathrm{SL}_n(\mathbb{R}) = \{X \in \mathrm{Mat}_{n \times n}(\mathbb{R}) : \mathrm{tr}(X) = 0\}$.

Problem 14 [TODO]

(a) Let $f_i : M \rightarrow N$, $i = 0, 1$, be two smooth maps between smooth manifolds M and N , and $f_i^* : \Omega^*(N) \rightarrow \Omega^*(M)$, $i = 0, 1$, be the induced chain maps between the respective de Rham complexes. Define the notion of chain homotopy between f_0^* and f_1^* . Here the co-boundary operators on the de Rham complexes are the exterior derivatives.

(b) Let X be a smooth vector field on a compact smooth manifold M , and let $\varphi_t : M \rightarrow M$ be the flow generated by X at time t , i.e. the solution of the differential equation $\frac{d\varphi_t}{dt}(x) = X(\varphi_t(x))$ with initial condition $\varphi_0(x) = x$. Find an explicit chain homotopy between the chain maps φ_0^* and φ_1^* , where φ_i^* , $i = 0, 1$, are the induced chain maps from $\Omega^*(M)$ to itself.

Hint: Use the formula that for any differential form ω and vector field X , the Lie derivative $\mathcal{L}_X \omega = d \circ i_X \omega + i_X \circ d\omega$. Here i_X is the contraction with respect to X .

Solution. (a): A chain homotopy between f_0^* and f_1^* is defined as follows: $\exists P : \Omega^n(N) \rightarrow \Omega^{n-1}(M)$ for all n such that $f_0^* - f_1^* = dP + Pd$.

(b):

Problem 15 [TODO]

Let $\omega = dx_1 \wedge dx_2 + dx_3 \wedge dx_4 + \dots + dx_{2n-1} \wedge dx_{2n}$ be a 2-form on $\mathbb{R}^{2n}$, where $(x_1, x_2, \dots, x_{2n})$ are the standard coordinates on $\mathbb{R}^{2n}$. Define an S^1 -action on $\mathbb{R}^{2n}$ as follows: for each $t \in S^1$, define $g_t : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ by considering $\mathbb{R}^{2n}$ as the direct sum of n copies of $\mathbb{R}^2$ and rotating each $\mathbb{R}^2$ summand an angle t . Let X be the vector field on $\mathbb{R}^{2n}$ defined by $X(x) = \frac{dg_t(x)}{dt}|_{t=0}$ for any $x \in \mathbb{R}^{2n}$.

- (a) Find the Lie derivative $\mathcal{L}_X \omega$ and a function f on $\mathbb{R}^{2n}$ such that $df = i_X \omega$.
- (b) The S^1 -action above induces an action on S^{2n-1} . Let $\mathbb{P}^{n-1}$ be the quotient space of S^{2n-1} by this S^1 -action. Show that the quotient space $\mathbb{P}^{n-1}$ has a natural smooth structure and that the tangent space of $\mathbb{P}^{n-1}$ at any point $\underline{x}$ can be identified with the quotient of the tangent space $T_x S^{2n-1}$ by the line spanned by $X(x)$, for any $x \in \underline{x}$. Here $\underline{x}$ is the orbit of x under the S^1 -action.
- (c) Show that ω descends to a well-defined 2-form on the quotient space $\mathbb{P}^{n-1}$ and that the 2-form so defined is closed.
- (d) Is the closed form in (c) exact? (*Hint:* For (c) and (d), use (a) and (b).)

*Solution.***Problem 16**

Suppose that $f : S^n \rightarrow S^n$ is a smooth map of degree not equal to $(-1)^{n+1}$. Show that f has a fixed point.

Solution. Assume the contrary, that f has no fixed point. First consider the following, $F : S^n \times [0, 1] \rightarrow \mathbb{R}^n$ where

$$F(x, t) = f(x)(1 - t) - xt.$$

Assume the contrary that $F(x, t) = 0$ for some x and t . Then $f(x)(1 - t) = xt$. If we take the norm of both sides

$$(1 - t) = t \Rightarrow t = \frac{1}{2}.$$

Thus $\frac{1}{2}f(x) = \frac{1}{2}x \Rightarrow f(x) = x$, meaning f has a fixed point and there is nothing to prove. Thus $F(x, t) \neq 0$ for all x and t . Consider the following homotopy then $H : S^n \times I \rightarrow S^n$ where

$$H(x, t) := \frac{F(x, t)}{\|F(x, t)\|}.$$

H is well defined. Note that

$$H(\cdot, 0) = \frac{F(x, 0)}{\|F(\cdot, 0)\|} = f(x)$$

$$H(x, 1) = -x.$$

Thus we have a homotopy between f and $-x$. Thus $\deg f = \deg([x \rightarrow -x]) = (-1)^{n+1}$. Contradicting our initial assumption on the degree of f .

Problem 17

- (a) Let G be a finitely presented group. Show that there is a topological space X with fundamental group $\pi_1(X) \cong G$.
 (b) Give an example of X in the case $G = \mathbb{Z} * \mathbb{Z}$, the free group on two generators.
 (c) How many connected, 2-sheeted covering spaces does the space X from (b) have?

Solution. (a): G is isomorphic to the following group by assumption

$$\langle x_1, \dots, x_n \mid r_1, \dots, r_k \rangle$$

Construct X as follows as a CW complex. You have a single 0 cell e_0 , 1 cells $e_1^1, \dots, e_1^n$, and 2 cells $e_2^1, \dots, e_2^k$. You attach the cells using the following attaching maps:

$$\begin{aligned} \partial e_1^i &\rightarrow e_0 \\ \partial e_2^i &\rightarrow r_i(e_1^1, \dots, e_1^n). \end{aligned}$$

Note the following $X^1 = \bigvee_{i=1}^n S^1$ hence $\pi_1(X) = \langle x_1, \dots, x_n \rangle$. Then adding the two cells with the specific attaching maps applies the corresponding relations, hence $\pi_1(X) = G$.

(b): $X = S^1 \vee S^1$

(c): Each connected 2-sheeted covering space corresponds to a index 2 subgroup of $\pi_1(X) = \mathbb{Z} * \mathbb{Z}$. Each index 2 subgroup corresponds to the kernel of some homomorphism from $\mathbb{Z} * \mathbb{Z} \rightarrow \mathbb{Z}_2$. These are the following homomorphisms from $\mathbb{Z} * \mathbb{Z} \rightarrow \mathbb{Z}_2$:

1. $a \rightarrow 1, b \rightarrow 1$: The kernel is $\langle a^2, b^2, ab \rangle$
2. $a \rightarrow 1, b \rightarrow 0$: The kernel is $\langle a^2, b \rangle$
3. $a \rightarrow 0, b \rightarrow 1$: The kernel is $\langle b^2, a \rangle$
4. $a \rightarrow 0, b \rightarrow 0$: The kernel is $\langle a, b \rangle$

Out of all of these, 1, 2, 3 are index 2 subgroups. Note that 2 and 3 produce isomorphic covering spaces. Thus up to isomorphism there are only 2 two sheeted coverings of X .

Problem 18 [TODO]

Let G be a connected topological group. Show that $\pi_1(G)$ is a commutative group.

Solution.

Problem 19

Show that if $\mathbb{R}^m$ and $\mathbb{R}^n$ are homeomorphic, then $m = n$.

Solution. Consider $\mathbb{R}^m/\{x\}$ which is homotopy equivalent to S^{m-1} and $\mathbb{R}^n/\{y\}$ which is equivalent to S^{n-1} . Then

$$\tilde{H}_k(S^{m-1}) = \delta(k, m-1)\mathbb{Z}$$

$$\tilde{H}_k(S^{n-1}) = \delta(k, n-1)\mathbb{Z}.$$

Suppose $m \neq n$. Then S^{m-1} and S^{n-1} have different homology groups hence are not homotopic. Thus $\mathbb{R}^m/\{x\}$ and $\mathbb{R}^n/\{y\}$ are not homotopic. Hence $\mathbb{R}^m$ and $\mathbb{R}^n$ cannot be homeomorphic.

Problem 20 [TODO]

Let N_g be the nonorientable surface of genus g , that is, the connected sum of g copies of $\mathbb{RP}^2$. Calculate the fundamental group and homology groups of N_g .

Solution. N_g has the following cell decomposition:

1. 0 cell: e_0
2. 1 cell: $e_1^1, \dots, e_1^g$. Where $\partial e_1^i \rightarrow e_0$.
3. 2 cells: e_2 . Where $\partial e_2 \rightarrow (e_1^1)^2 \dots (e_1^g)^2$

Thus

$$\pi_1(N_g) = \langle a_1, \dots, a_g \mid a_1^2 \dots a_g^2 \rangle$$

We have the following Cellular Chain complex:

$$0 \rightarrow \mathbb{Z}\langle e_2 \rangle \xrightarrow{\begin{pmatrix} 2 \\ \vdots \\ 2 \end{pmatrix}} \mathbb{Z}\langle e_1^1, \dots, e_1^g \rangle \xrightarrow{0} \mathbb{Z}\langle e_0 \rangle \rightarrow 0$$

Thus

$$H_0(N_g) = \mathbb{Z}\langle e_0 \rangle$$

$$H_1(N_g) = \frac{\mathbb{Z}\langle e_1^1, \dots, e_1^g \rangle}{\mathbb{Z}\langle 2e_1^1, \dots, 2e_1^g \rangle} = \mathbb{Z}_2\langle e_1^1, \dots, e_1^g \rangle$$

$$H_2(N_g) = 0$$

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Problem 21

Let M_n be the space of all $n \times n$ matrices with real entries and let S_n be the subset consisting of all symmetric matrices. Consider the map $F : M_n \rightarrow S_n$ defined by $F(A) = AA^t - I$, where I is the identity matrix and A^t is the transpose of A .

- (a) Show that $0_{n \times n}$ (the $n \times n$ matrix with all entries 0) is a regular value of F .
- (b) Deduce that $O(n)$, the set of all $n \times n$ matrices such that $A^{-1} = A^t$, is a submanifold of M_n .
- (c) Find the dimension of $O(n)$ and determine the tangent space of $O(n)$ at the identity matrix as a subspace of the tangent space of M_n , which is M_n itself.

Solution. (a): Let $E_{i,j} = [\delta(a,i)\delta(b,j)]_{1 \leq a,b \leq n}$ be the elementary matrices. $F^{-1}(\{0_{n \times n}\}) := \{M \in M_n : M^{-1} = M^T\} = O(n)$. Fix $M = [m_{i,j}]_{0 \leq i,j \leq 1} \in O(n)$. For $E_{i,j} \in M_n$. Let $\varphi_{i,j}(t) = M + tE_{i,j}$.

$$\begin{aligned} F(\varphi_{i,j}(t)) &= (M + tE_{i,j})(M + tE_{i,j})^T - I \\ &= F(M) + t(ME_{j,i} + E_{i,j}M^T) + O(t^2) \\ &= t(ME_{j,i} + E_{i,j}M^T) + O(t^2) \\ &= t\left(\sum_{k=1}^n m_{k,j}E_{k,i} + \sum_{k=1}^n m_{k,j}E_{i,k}\right) + O(t^2) \\ &= t\left(\sum_{k=1}^n m_{k,j}(E_{k,i} + E_{i,k})\right) + O(t^2) \end{aligned}$$

Thus

$$dF_M(E_{i,j}) = \sum_{k=1}^n m_{k,j}(E_{k,i} + E_{i,k})$$

for all $1 \leq i, j \leq n$.

$$\begin{aligned} dF_M\left(\sum_{j=1}^n m_{l,j}E_{i,j}\right) &= \sum_{j=1}^n m_{l,j}dF_M(E_{i,j}) \\ &= \sum_{j=1}^n \sum_{k=1}^n m_{l,j}m_{k,j}(E_{i,k} + E_{k,i}) \\ &= \sum_{l=1}^n \sum_{k=1}^n \delta(l,k)(E_{i,k} + E_{k,i}) = E_{i,l} + E_{l,i} \end{aligned}$$

For all $1 \leq i, l \leq n$. Hence dF_M is surjective and M is a regular point. Hence $0_{n \times n}$ is a regular value.

(b): By the regular value theorem $O(n)$ is a submanifold of M_n .

(c):

$$\dim O(n) = \dim T_I O(n).$$

Note the fact that

$$T_I O(n) = \ker dF_I$$

$$\begin{aligned} F(I + tX) &= (I + tX)(I + tX)^T - I \\ &= t(X^T + X) + t^2(\mathbb{X}^T) \end{aligned}$$

Thus

$$dF_I(X) = X^T + X.$$

If $X \in \ker dF_I \Leftrightarrow X^T + X = 0 \Leftrightarrow X = -X^T$ ie $X \in \ker dF_I \Leftrightarrow X$ skew symmetric. Skew Symmetric matrices have dimension

$$n - 1 + n - 2 + \dots + 1 = \frac{n(n-1)}{2}.$$

Thus $O(n)$ is a dimension $0.5n(n-1)$ manifold.

Problem 22 [TODO]

Show that $T^2 \times S^n$, $n \geq 1$, is parallelizable, where S^n is the n -sphere, $T^2 = S^1 \times S^1$ is the two torus, and a manifold of dimension k is said to be parallelizable if there are k vector fields $V_1, \dots, V_k$ on it with $V_1(p), \dots, V_k(p)$ linearly independent for all points p of the manifold.

Solution. Note the following

$$T(T^2 \times S^n) \cong T^2 \times \mathbb{R} \times \mathbb{R} \times T(S^n)$$

Since $T(T^2)$ is parallelizable (use ∂_x, ∂_y as vector fields). Note the following that $\mathbb{R} \times S^n \cong \mathbb{R}^{n+1}/\{0\}$ where $(t, x) \rightarrow e^t x$. Thus $\mathbb{R}^2 \times T(S^n) = T(\mathbb{R} \times S^n) = T(\mathbb{R}^{n+1}/\{0\}) = (\mathbb{R}^{n+1}/\{0\}) \times \mathbb{R}^{n+1}$ where our vector fields are $\partial_{z_1}, \dots, \partial_{z_{n+1}}$ where $(z_1, \dots, z_{n+1})$ are the coordinates of $\mathbb{R}^{n+1}$. Pull back the vector fields using the diffeomorphism to get the $(n+1)$ linearly independent vectors of $T(\mathbb{R} \times S^n)$. We have $T^2 \times T(\mathbb{R} \times S^n) = T^2 \times (\mathbb{R}^{n+1}/\{0\}) \times \mathbb{R}^{n+1} = T^2 \times \mathbb{R} \times S^n \times \mathbb{R}^{n+1} = T^2 \times S^n \times \mathbb{R}^{n+2}$ where the second last equality comes from the diffeomorphism.

Problem 23 [TODO]

Suppose $\pi : M_1 \rightarrow M_2$ is a C^∞ map of one connected differentiable manifold to another. And suppose for each $p \in M_1$, the differential $\pi_* : T_p M_1 \rightarrow T_{\pi(p)} M_2$ is a vector space isomorphism.

- Show that if M_1 is connected, then π is a covering space projection.
- Give an example where M_2 is compact but $\pi : M_1 \rightarrow M_2$ is not a covering space (but has the π_* isomorphism property).

Solution.

Problem 24 [TODO]

Let $\mathcal{F}^k(M)$ denote the differentiable (C^∞) k -forms on a manifold M . Suppose U and V are open subsets of a differentiable manifold.

(a) Explain carefully how the usual exact sequence

$$0 \rightarrow \mathcal{F}(U \cup V) \rightarrow \mathcal{F}(U) \oplus \mathcal{F}(V) \rightarrow \mathcal{F}(U \cap V) \rightarrow 0$$

arises.

(b) Write down the “long exact sequence” in de Rham cohomology associated to the short exact sequence in part (a) and describe explicitly how the map

$$H_{deR}^k(U \cap V) \rightarrow H_{deR}^{k+1}(U \cup V)$$

arises.

Solution.

Problem 25 [TODO]

Explain carefully why the following holds: if $\pi : S^N \rightarrow M$, $N > 1$, is a covering space with M orientable, then every closed k -form on M , $1 \leq k < N$, is exact. (Suggestion: Recall that the covering transformations in this situation form a group G with $S^N/G \cong M$.)

Solution.

Problem 26

Calculate the singular homology of $\mathbb{R}^n$, $n > 1$, with k points removed, $k \geq 1$. (Your answer will depend on k and n .)

Solution. $X = \mathbb{R}^n / \{x_1, \dots, x_k\}$ deformation retracts to a $\bigvee_{i=1}^k S^{n-1}$.

$$H_m(X) \cong H_m\left(\bigvee_{i=1}^k S^{n-1}\right) \cong \bigoplus_{i=1}^k H_m(S^{n-1}) = \begin{cases} \mathbb{Z}^k & \text{if } m = n-1 \\ 0 & \text{otherwise} \end{cases}$$

for $m > 0$. For $H_0(X) = \mathbb{Z}$, since X connected.

Problem 27 [TODO]

(a) Explain what is meant by adding a handle to a 2-sphere, for a two dimensional orientable surface in general.

(b) Show that a 2-sphere with a positive number of handles attached cannot be simply connected.

Solution.

Problem 28 [TODO]

FIX ANSWER!

- (a) Define the degree $\deg f$ of a C^∞ map $f : S^2 \rightarrow S^2$ and prove that $\deg f$ as you present it is well-defined and independent of any choices you need to make in your definition.
- (b) Prove in detail that for each integer k (possibly negative), there is a C^∞ map $f : S^2 \rightarrow S^2$ of degree k .

Solution. (a): S^2 as a CW complex is a 0 cell e_0 , and a 2 cell e_2 (whose boundary maps to the 0 cell). $H_0(S^2) = \mathbb{Z}\langle e_0 \rangle$ and $H_2(S^2) = \mathbb{Z}\langle e_2 \rangle$ and $H_1(S^2) = 0$. Thus any map $f : S^2 \rightarrow S^2$ induces a map on homology, and since in both the domain and range $H_2(S^2)$ has a unique generator e_2 . $f(e_2) = \lambda e_2$. $\deg f = \lambda$.

(b): Let $f_0 : S^2 \rightarrow S^2$ where $S^2 \rightarrow (1, 0, 0)$. Because $f_0(S^2) \subset \frac{S^2}{-1, 0, 0}$ which deformation retracts to $(1, 0, 0)$ and hence $H_2(f_0(S^2)) = 0$ ie $f_0(e_2) = 0$ and $\deg(f_0) = 0$. Note that the antipodal map $A(x) = -x$ has degree -1 . Note that $\deg(g \circ f) = \deg g \deg f$. Thus it suffices to find a map $g_k : S^2 \rightarrow S^2$ such that $\deg g_k = k$ for all $k \in \mathbb{N}$. Identify S^2 with $\mathbb{C}P^1$. Consider the map $g_k(z) = z^k$. By local degree, $g_k^{-1}(\{1\})$ are the k roots of unity and g_k is orientation perveing. Hence $\deg g_k = k$.

Problem 29 [TODO]

Explain how Stokes' Theorem for manifolds with boundary gives, as a special case, the classical divergence theorem (about $\iiint_U \operatorname{div} V, d(\operatorname{vol})$), where U is a bounded open set in $\mathbb{R}^3$ with smooth boundary and V is a C^∞ vector field on $\mathbb{R}^3$.

Solution.

Problem 30 [TODO]

- (a) Show that every map $F : S^n \rightarrow S^1 \times \cdots \times S^1$ (k copies of S^1) is null-homotopic (homotopic to a constant map).
- (b) Show that there is a map $F : S^1 \times \cdots \times S^1$ (n copies) $\rightarrow S^n$ such that F is not null-homotopic.
- (c) Show that every map $F : S^n \rightarrow S^{n_1} \times S^{n_2} \times \cdots \times S^{n_k}$, $n_1 + \cdots + n_k = n$, $n_j > 0$, $k \geq 2$, has degree 0. (You may use any definition of degree you like, and you may assume F is C^∞ .)

Solution. a: For $n > 1$: $\mathbb{T}^k = \mathbb{R}^k / \mathbb{Z}^k$. Thus $\pi_1(\mathbb{T}^k) = \mathbb{Z}^k$. $F_* : 0 = \pi_1(S^n) \rightarrow \pi_1(\mathbb{T}^k) = \mathbb{Z}^k$. By the lifting criterion $\tilde{F} : S^n \rightarrow \mathbb{R}^k$ exists where $F = p \circ \tilde{F}$. Consider the straight line homotopy $\tilde{H}(x, t) = (1 - t)\tilde{F}(x)$. Then let $H = p \circ \tilde{H}$. Then $H(0, x) = p((1 - 0)\tilde{F}(x)) = F(x)$, $H(1, x) = p((1 - x)\tilde{F}(x)) = p(0) = [0]$. Thus F is null-homotopic.

(b):

(c):

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Problem 31 [TODO]

Let M be an (abstract) compact smooth manifold. Prove that there exists some $n \in \mathbb{Z}^+$ such that M can be smoothly embedded in the Euclidean space $\mathbb{R}^n$.

Solution.

Problem 32 [TODO]

Prove that the real projective space $\mathbb{RP}^n$ is a smooth manifold of dimension n .

FIX PROOF

Solution.

$$\mathbb{RP}^n = \{[x_1 : \dots : x_{n+1}] : (x_1, \dots, x_n, x_{n+1}) \neq 0, [x_1 : \dots : x_{n+1}] \sim [kx_1 : \dots : kx_{n+1}]\}.$$

Define

$$U_i := \{[x_1 : \dots : x_{i-1} : 1 : x_{i+1} : \dots : x_{n+1}] : (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{n+1}) \in \mathbb{R}^n\}$$

Thus U_i is homeomorphic $\mathbb{R}^n$ where

$$\begin{aligned} \varphi_i : \mathbb{R}^n &\rightarrow U_i \\ (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{n+1}) &\rightarrow [x_1 : \dots : x_{i-1} : 1 : x_{i+1} : \dots : x_{n+1}] \end{aligned}$$

is the diffeomorphism.

$$U_i \cap U_j = \{[x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_{j-1}, 1, x_{j+1}, \dots, x_{n+1}] : (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{j-1}, x_{j+1}, \dots, x_{n+1}) \in \mathbb{R}^{n-1}\}$$

Note that $\varphi_i^{-1}(U_i \cap U_j) = \mathbb{R}^{j-1} \times (\mathbb{R} - \{0\}) \times \mathbb{R}^{n-j-1}$

$$\begin{aligned} (\varphi_j^{-1} \circ \varphi_i)(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) &= \varphi_j^{-1}[x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_j, \dots, x_n] \\ &= \varphi_j^{-1}\left(\left[\frac{x_1}{x_j}, \dots, \frac{x_{j-1}}{x_j}, 1, \frac{x_{j+1}}{x_j}, \dots, \frac{x_n}{x_j}\right]\right) \\ &= \left(\frac{x_1}{x_j}, \dots, \frac{1}{x_j}, \dots, \frac{x_{j-1}}{x_j}, \frac{x_{j+1}}{x_j}, \dots, \frac{x_n}{x_j}\right) \end{aligned}$$

This is smooth. So the transition functions are smooth. Hence $\mathbb{RP}^n$ is a smooth manifold of dimension n .

Problem 33 [TODO]

Let M be a compact, simply connected smooth manifold of dimension n . Prove that there is no smooth immersion $f : M \rightarrow T^n$, where $T^n = S^1 \times \cdots \times S^1$ is the n -torus.

Solution.

Problem 34 [TODO]

Give a topological proof of the Fundamental Theorem of Algebra: any non-constant single-variable polynomial with complex coefficients has at least one complex root.

Solution.

Problem 35 [TODO]

Let $f : M \rightarrow N$ be a smooth map between two manifolds M and N . Let α be a p -form on N . Show that $d(f^*\alpha) = f^*(d\alpha)$.

Solution. Let $g \in \Omega^0(N)$ be a 0 form. Let X be a vector field

$$\begin{aligned} d(f^*(g))(X) &= d(g \circ f)(X) \\ &= X(g \circ f) \\ f^*(dg)[X] &= dg(f_*X) \\ &= (f_*X)[g] \\ &= X(g \circ f) \end{aligned}$$

Since this is true for all vector fields X we have that $d(f^*(g)) = f^*(dg)$.

Assume this is true for forms $n = 1, \dots, k$. We want to prove this is true for a $k + 1$ form. ν is a k form and $g \in \Omega^0(N)$.

$$\begin{aligned} d(f^*(g \wedge \nu)) &= d((f^*g) \wedge (f^*\nu)) \\ &= d(f^*g) \wedge f^*\nu + f^*g \wedge d(f^*\nu) \\ &= f^*(dg) \wedge f^*\nu + f^*g \wedge f^*(d\nu) \\ &= f^*(dg \wedge \nu + g \wedge d\nu) \\ f^*(d(g \wedge \nu)) &= f^*(dg \wedge \nu + g \wedge d\nu) \end{aligned}$$

For all $\alpha \in \Omega^{k+1}(N)$ is a linear combination of $g \wedge \nu$ and since both d and f are linear it is true for all $\Omega^{k+1}(N)$.

Thus by induction this is true for all p -forms on N .

Problem 36 [TODO]

- (a) What are the de Rham cohomology groups of a smooth manifold?
- (b) State de Rham's theorem.

Solution.

Problem 37 [TODO]

Consider the form $\omega = (x^2 + x + y) dy \wedge dz$ on $\mathbb{R}^3$. Let $S^2 = \{x^2 + y^2 + z^2 = 1\} \subset \mathbb{R}^3$ be the unit sphere, and $i : S^2 \rightarrow \mathbb{R}^3$ the inclusion.

- (a) Calculate $\int_{S^2} \omega$.
- (b) Construct a closed form α on $\mathbb{R}^3$ such that $i^* \alpha = i^* \omega$, or show that such a form α does not exist.

Solution.

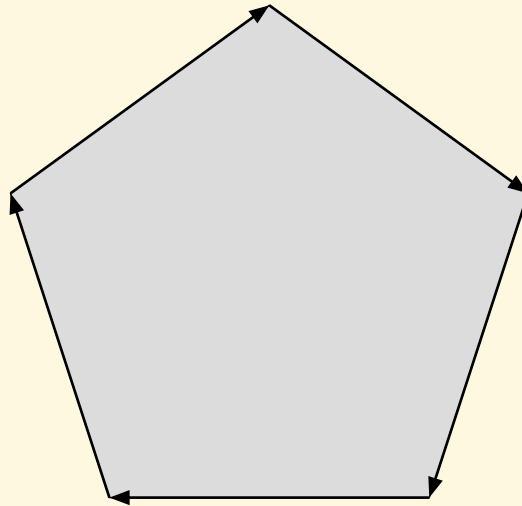
Problem 38 [TODO]

- (a) Let M be a Möbius band. Using homology, show that there is no retraction from M to ∂M .
- (b) Let K be a Klein bottle. Show that there exist homotopically nontrivial simple closed curves γ_1 and γ_2 on K such that K retracts to γ_1 , but does not retract to γ_2 .

Solution.

Problem 39 [TODO]

Let X be the topological space obtained from a pentagon by identifying its edges as in the picture:



Calculate the homology and cohomology groups of X with integer coefficients.

Solution.

Problem 40 [TODO]

Let X, Y be topological spaces and $f, g : X \rightarrow Y$ two continuous maps. Consider the space Z obtained from the disjoint union $Y \sqcup (X \times [0, 1])$ by identifying $(x, 0) \sim f(x)$ and $(x, 1) \sim g(x)$ for all $x \in X$. Show that there is a long exact sequence of the form:

$$\dots \rightarrow H_n(X) \rightarrow H_n(Y) \rightarrow H_n(Z) \rightarrow H_{n-1}(X) \rightarrow \dots$$

Solution.